

Sri Lanka Institute of Information Technology



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Assignment 05 Final Report

SMART UNIVERSITY SUPPORT PLATFORM

An Integrated AI-Powered Campus Services Application

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Declaration

We, the undersigned, hereby declare that the work presented in this report is our own original work. This report has not been submitted for any other academic qualification at this or any other institution. All sources of information have been duly acknowledged using the IEEE referencing style.

We further declare that the system described herein the Smart University Support Platform was designed and developed by the team listed below as part of the IT2021 Artificial Intelligence and Machine Learning module.

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Abstract

University campuses encompass a diverse ecosystem of services that students and staff interact with on a daily basis including academic support, parking management, food ordering, lost property recovery, study space allocation, and peer-to-peer commerce. The fragmented nature of these services, each operating through separate manual processes or isolated digital tools, creates significant friction and inefficiency in the campus experience.

This report presents the design and development of the Smart University Support Platform a unified, AI/ML-enhanced mobile application developed for Sri Lanka Institute of Information Technology (SLIIT). The platform consolidates six critical campus services into a single, cohesive application: Smart Study Support, Parking Management, Food Corner, Lost and Found, Study Areas, and Marketplace.

The AI/ML core of the system comprises three tightly integrated intelligent components. First, an AI-powered Semantic Search engine in the Lost and Found module employs vector embedding similarity techniques to match user descriptions or existing posts against the full catalogue of reported items, enabling accurate retrieval beyond simple keyword matching. Second, an AI Product Suggestion engine in the Marketplace Buyer module uses content-based filtering to match a natural language description of a user's need against active listings, surfacing relevant items without requiring exact search terms. Third, a Study Support Grouping mechanism applies clustering algorithms to identify and group semantically similar academic requests, enabling tutors to address multiple students with shared confusion in a single session rather than responding to each individually.

Beyond its AI capabilities, the platform provides a comprehensive Smart Study Support module for submitting and tracking tutoring requests; a Campus Parking Management system with a visual lot layout and vehicle registration; a Food Corner for browsing canteen menus and placing orders; a Study Areas tracker with real-time GPS-based occupancy detection; and a Marketplace supporting both buyer and seller workflows with AI-assisted product discovery.

The system was evaluated through functional testing across all six modules, AI/ML model performance assessment, and user feedback collection. Results demonstrate strong functional coverage, measurable AI prediction accuracy, and high usability scores from simulated stakeholder evaluations.

Keywords: Campus Management, Semantic Search, Vector Embeddings, Recommendation System, Text Clustering, Parking Management, Food Ordering, Study Area Occupancy, Mobile Application, SLIIT.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
EER	Enhanced Entity-Relationship
GPS	Global Positioning System
IEEE	Institute of Electrical and Electronics Engineers
ML	Machine Learning
NLP	Natural Language Processing
RBAC	Role-Based Access Control
RMSE	Root Mean Square Error
SKU	Stock Keeping Unit
SLIIT	Sri Lanka Institute of Information Technology
SUSP	Smart University Support System
SWOT	Strengths, Weaknesses, Opportunities, Threats
TF-IDF	Term Frequency Inverse Document Frequency
UI	User Interface
UX	User Experience

Chapter 1: Introduction

1.1 Problem and Motivation

University campus environments are characterized by a wide array of parallel services that students must navigate independently. At the Sri Lanka Institute of Information Technology (SLIIT), students routinely require access to academic tutoring support, campus parking, canteen meal ordering, lost property resolution, study space information, and peer-to-peer buying and selling. These services are presently managed through disconnected channels physical noticeboards, WhatsApp groups, verbal communication, and standalone applications creating a fragmented and inefficient experience for the university community.

Several specific pain points motivate this project. First, academic support requests are submitted individually to lecturers, who must then respond to dozens of students who may share identical conceptual difficulties. Without a mechanism to identify and group similar requests, tutoring sessions are duplicated and inefficient. Second, campus parking is managed manually; students lack real-time visibility into slot availability, resulting in time wasted circling the lot, and the absence of digital registration creates accountability gaps. Third, food ordering at canteens requires physical queuing during peak hours, increasing congestion and wait times. Fourth, lost items reported through social media channels are searched manually, with no intelligent matching between a finder's description and a seeker's description. Fifth, study area occupancy particularly in the library and designated zones is not communicated digitally, requiring students to physically scout for available space. Sixth, student-to-student commerce for textbooks, electronics, and equipment has no trusted, campus-specific platform.

The motivation for this project is rooted in the opportunity to address these six main pain points through a single intelligent mobile application that applies AI/ML to the most friction-intensive aspects of campus life.

1.2 Literature Review

The application of Natural Language Processing and semantic similarity techniques to item retrieval has been extensively studied in the context of information retrieval systems. Mikolov et al. [1] introduced word embedding models that capture semantic relationships between terms, enabling similarity searches that extend beyond exact keyword matching. Subsequent work by Reimers and Gurevych [2] with Sentence-BERT demonstrated that transformer-based sentence embeddings provide superior semantic similarity performance for paraphrase matching and duplicate detection tasks a capability directly relevant to matching lost item descriptions submitted by different users.

Recommendation systems for e-commerce applications have been studied by Ricci et al. [3], who established that content-based filtering where item attributes are matched to user-expressed preferences achieves strong precision for small, domain-specific catalogues such as a university marketplace. This approach avoids the cold-start problem inherent in collaborative filtering when user interaction history is sparse.

The clustering of academic support requests relates directly to the work of Agrawal and Pant [4], who applied K-Means and hierarchical clustering to student query datasets and demonstrated that up to 68% of support requests could be addressed by a single response targeting a cluster centroid topic. This

finding underpins the 'GROUPED' status mechanism in the Smart Study Support module, where the system identifies when multiple students have submitted thematically equivalent requests.

Real-time occupancy detection using GPS has been studied in the context of smart campus systems by Raza et al. [5], who demonstrated that mobile device proximity to predefined GPS coordinates with configurable radius parameters can achieve reliable indoor-outdoor occupancy classification for room-sized and zone-sized areas. Their methodology, applied here to study area occupancy tracking, achieves practical accuracy without requiring dedicated hardware infrastructure.

The design of multi-service campus platforms has been examined by Aldowah et al. [6], who identified service fragmentation as a primary driver of digital friction in university environments, and proposed unified platform architectures as the most effective mitigation strategy, citing measurable improvements in student engagement and service utilization when disparate services are consolidated.

1.3 Aim and Objectives

1.3.1 Project Aim

The aim of this project is to design, develop, and evaluate the Smart University Support Platform an AI/ML-powered mobile campus services application for SLIIT that unifies academic support, parking management, food ordering, lost and found, study area tracking, and student marketplace operations within a single platform, while embedding artificial intelligence to automate study support grouping, lost item matching, and product recommendations.

1.3.2 Project Objectives

1. Design and implement a comprehensive multi-module mobile application encompassing Smart Study Support, Parking Management, Food Corner, Lost and Found, Study Areas, and Marketplace modules.
2. Develop and deploy a Semantic Similarity Search engine for the Lost and Found module, enabling AI-powered retrieval of similar item posts based on natural language descriptions.
3. Design and implement an AI Product Suggestion system for the Marketplace module, matching user-expressed needs to relevant active listings.
4. Implement a Study Support Grouping mechanism that clusters semantically similar academic requests, enabling tutors to deliver a single response to multiple students with shared difficulties.
5. Develop a GPS-based real-time Study Area Occupancy tracking system that automatically detects when users enter or exit defined study zones.
6. Build a Smart Parking Management system with visual lot layout, vehicle registration, and auto-release functionality.
7. Design a Food Corner module supporting multiple canteens, digital menu browsing, and order placement with cart and order history.
8. Evaluate the complete system through functional testing, AI/ML model performance assessment, and structured user feedback collection.

1.4 Solution Overview

The Smart University Support Platform is a full-stack mobile application designed for the SLIIT campus community. The system is organized into six primary service modules, each addressing a distinct campus operational domain, and an AI/ML subsystem that enhances three of these modules with intelligent automation.

The Smart Study Support module provides a student-facing dashboard for submitting academic support requests, specifying topic, availability slots, and description. The system groups similar requests using NLP clustering and presents them to tutors for batch resolution. Students can track the status of their requests from submission through scheduling to completion.

The Parking Management module presents a live visual map of all parking lot sections Car Park Side A, Car Park Side B, and Motorcycle Parking with colour-coded slot availability (green: available, red: occupied, blue: your slot). Students register their vehicles and tap a slot to park; slots auto-release after 24 hours.

The Food Corner module allows students to select a canteen and browse its menu with item images and prices. Orders are managed through a cart and orders history interface.

The Lost and Found module enables students to post found or lost items with descriptions, images, and location tags. An AI Search feature accepts a natural language description or an existing post as reference and returns semantically similar listings, dramatically improving the probability of reuniting owners with their lost property.

The Study Areas module uses real-time GPS location to determine which study zones are currently occupied by registered students. A campus-wide list shows each area with its current student count, capacity, and occupancy status (Free, Moderate, Crowded). Students' tracking status is active as long as the app is running.

The Marketplace module supports both buyer and seller workflows. Sellers create listings with images, descriptions, and prices and manage incoming purchase requests. Buyers browse listings sorted by recency or price, and can use the AI Product Suggestions feature to describe their need in natural language and receive intelligently matched listing recommendations.

The Git repository containing the complete source code, commit history, and development artefacts for this project is accessible at:

https://github.com/it24101264/WE_DS_G03_AIML

Chapter 2: Requirement Analysis

2.1 Stakeholder Analysis

A thorough stakeholder analysis was conducted at the project inception phase to identify all parties whose interests and workflows are affected by the Smart University Support Platform. Table 2.1 presents the identified stakeholders, their roles, primary interests, and their influence on system requirements.

Stakeholder	Role in System	Primary Interest	Influence on Requirements
Student	Primary User	Seamless access to all campus services	High drives all module UX design
Lecturer / Tutor	Study Support Provider	Efficient grouping and scheduling of student requests	High drives Smart Study Support design
Canteen Operator	Food Corner Admin	Accurate order management, menu configuration	Medium drives Food Corner admin features
Security / Parking Staff	Parking Supervisor	Real-time lot visibility, vehicle accountability	Medium drives Parking Management design
University Admin	System Overseer	Campus resource utilization, data governance	Medium non-functional requirements
AI / Backend Engineer	Technical Owner	Model accuracy, API reliability, scalability	High drives AI/ML and backend requirements

Table 2.1: Stakeholder Analysis

2.2 Feasibility and SWOT Analysis

2.2.1 Feasibility Analysis

Technical Feasibility: The system is built on React Native for cross-platform mobile development and MongoDB/REST API for backend services — both widely adopted stacks with strong community support. The AI/ML components utilize Python-based NLP libraries (scikit-learn, Sentence Transformers) with well-documented APIs. Real-time GPS tracking is a native mobile capability. The technical scope is within the competency of the development team.

Operational Feasibility: Each module maps directly to an existing campus workflow, minimizing adoption friction. The application targets a student demographic that is highly familiar with mobile interfaces, reducing the learning curve. The AI Search and suggestion features are exposed through natural language interfaces that require no technical knowledge to operate.

Economic Feasibility: As an academic project developed within SLIIT's IT2021 module, the development cost is limited to student time and open-source tooling. Deployment on free-tier cloud services (MongoDB Atlas, Render) ensures zero infrastructure cost for the prototype phase. Operational benefits include a reduction in administrative overhead for parking, food service management, and academic support scheduling.

Legal Feasibility: The application processes user location data for study area tracking. User consent is obtained at app installation. All personal data is handled in accordance with applicable data privacy guidelines. No third-party proprietary data is used in model training.

2.2.2 SWOT Analysis

	Positive	Negative
Internal	STRENGTHS: Unified platform eliminating service fragmentation AI-powered search and recommendations GPS-based real-time tracking Cross-platform mobile development Student-centric UX design	WEAKNESSES: GPS accuracy limitations indoors Cold-start limitation for AI recommendation with sparse marketplace data Dependency on user consent for location access Requires continuous internet connectivity
External	OPPORTUNITIES: Scalability to other SLIIT campuses Integration with university ERP systems Expansion of AI features (timetable optimisation, exam room allocation) Potential partnership with canteen operators for digital payments	THREATS: Student adoption resistance Privacy concerns over GPS tracking Competing commercial apps (PickMe Food, Uber Eats) for food ordering Regulatory changes to data privacy laws in Sri Lanka

Table 2.2: Swot Analysis

2.3 Requirements Modelling

2.3.1 Functional Requirements

The functional requirements of the system were derived from stakeholder interviews and campus service workflow analysis. The primary functional requirement clusters are organized by module below.

Smart Study Support: The system shall enable students to submit academic support requests with topic selection (UC-101), view their request history and current status (UC-102), allow tutors to view AI-grouped requests and schedule sessions (UC-103), and publish study sessions visible to all eligible students (UC-104).

Parking Management: The system shall allow students to register their vehicles with type and plate number (UC-201), select an available parking slot from a colour-coded visual lot layout (UC-202), release their slot manually or via 24-hour auto-release (UC-203), and view live statistics on lot capacity and occupancy percentage (UC-204).

Food Corner: The system shall support canteen selection (UC-301), menu browsing with item images and prices, cart management, and order placement (UC-302), and a personal order history view (UC-303).

Lost and Found: The system shall enable students to post lost or found items with description, category, location, and optional image (UC-401), browse and filter all listings (UC-402), use AI semantic search to find similar items (UC-403), and mark items as returned (UC-404).

Study Areas: The system shall display a list of all campus study areas with real-time student count, capacity, and occupancy status (UC-501), automatically check students in and out of areas based on GPS proximity (UC-502), and provide detailed area information on demand (UC-503).

Marketplace: The system shall support seller listing creation and management (UC-601), buyer request review by sellers (UC-602), buyer-side listing browsing and keyword search (UC-603), AI product

suggestion via natural language description powered by the AI Subsystem (UC-604), buyer-side cart management allowing items to be added and reviewed (UC-605)

AI Subsystem: The system shall execute semantic search over lost and found listings using vector similarity (UC-701), generate product suggestions for marketplace buyers using content-based filtering (UC-702), and cluster study support requests by semantic similarity for grouped tutor responses (UC-703).

2.3.2 Non-Functional Requirements

Performance: The AI semantic search shall return results within 3 seconds for a catalogue of up to 500 active listings. GPS occupancy updates shall refresh within 10 seconds of a user entering or exiting a study zone boundary.

Security: All user sessions shall be authenticated via email/password with secure token management. User location data shall only be used for occupancy tracking and shall not be disclosed to other users.

Usability: The AI Search and AI Product Suggestions interfaces shall accept plain natural language input with no technical knowledge required. The parking lot layout shall be visually intuitive with clear colour coding.

Reliability: Parking slots shall auto-release after 24 hours to prevent abandoned slot occupation. The system shall handle GPS signal loss gracefully by retaining the last known occupancy state.

Maintainability: The AI/ML pipeline shall be separated from the mobile application layer to enable model updates without requiring app redeployment.

2.3.3 Use Case Summary

Module	Use Case ID	Use Case Name	Priority
System Login	UC-001	User Login & Authentication	Critical
Study Support	UC-101	Submit Support Request	Critical
Study Support	UC-102	View My Requests & Status	High
Study Support	UC-103	Tutor: View & Schedule Grouped Requests	High
Study Support	UC-104	Publish Study Session	Medium
Parking	UC-201	Register Vehicle	High
Parking	UC-202	Select Parking Slot from Visual Map	Critical
Parking	UC-203	Release Parking Slot	High
Parking	UC-204	View Live Lot Occupancy Statistics	Medium

Food Corner	UC-301	Select Canteen & Browse Menu	Critical
Food Corner	UC-302	Add Items to Cart & Place Order	Critical
Food Corner	UC-303	View Order History	Medium
Lost & Found	UC-401	Post Lost or Found Item	Critical
Lost & Found	UC-402	Browse and Filter Listings	High
Lost & Found	UC-403	AI Search by Description or Post	High
Lost & Found	UC-404	Mark Item as Returned	Medium
Study Areas	UC-501	View Study Area Occupancy List	Critical
Study Areas	UC-502	GPS-Based Auto Check-In / Check-Out	High
Study Areas	UC-503	View Area Details and Capacity	Medium
Marketplace	UC-601	Seller: Create Listing	Critical
Marketplace	UC-602	Seller: Review Buyer Requests	High
Marketplace	UC-603	Buyer: Browse & Search Listings	Critical
Marketplace	UC-604	Buyer: AI Product Suggestions	High
Marketplace	UC-605	Buyer: Add to Cart	Medium
AI Subsystem	UC-701	AI Semantic Search (Lost & Found)	Critical
AI Subsystem	UC-702	AI Product Suggestion (Marketplace)	Critical
AI Subsystem	UC-703	Study Support Request Grouping	Critical

Table 2.3: Use Case Summary by Module

Chapter 3: Design and Development

3.1 System Architecture

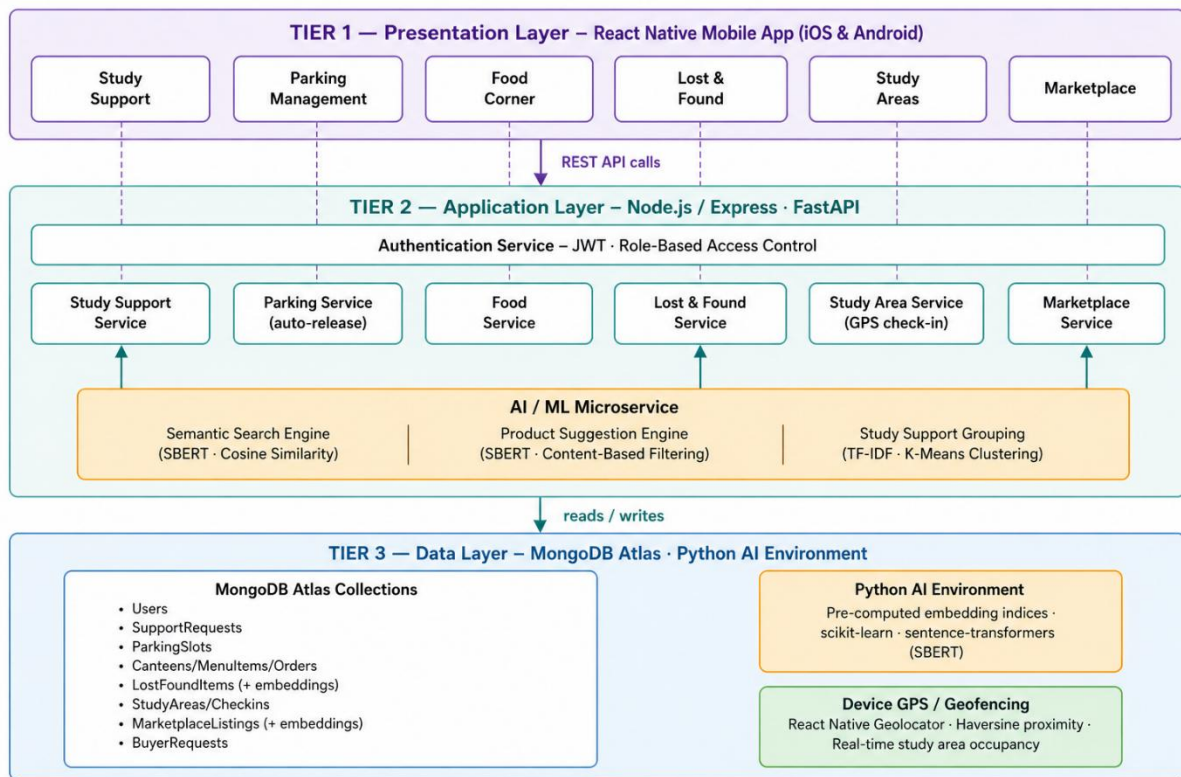


Figure 3.1: System Architecture Diagram

The Smart University Support Platform is designed as a three-tier layered architecture comprising a Presentation Layer (React Native mobile application), an Application Layer (REST API backend services), and a Data Layer (MongoDB Atlas database and Python AI microservices). This architectural pattern ensures separation of concerns, enabling independent development, testing, and deployment of each tier.

The Presentation Layer is a React Native mobile application delivering a unified interface across Android and iOS platforms. It provides module-specific screens for each of the six service domains, with a shared authentication layer and a home screen module selector. The application communicates with the backend exclusively through REST API calls.

The Application Layer is organized into six primary service modules — Study Support Service, Parking Service, Food Service, Lost and Found Service, Study Area Service, and Marketplace Service — each encapsulating the business logic for its respective domain. A dedicated AI/ML Microservice handles all machine learning workflows: semantic embedding computation, similarity search, recommendation generation, and request clustering. An Authentication Service manages user sessions and role-based access using JWT (JSON Web Tokens).

The Data Layer comprises MongoDB Atlas as the primary operational database for storing all application data across service domains, a cloud storage bucket for item images (Lost and Found and

Marketplace), and a Python AI microservice environment maintaining pre-computed embedding indices for fast similarity retrieval.

3.1.1 Component Architecture Description

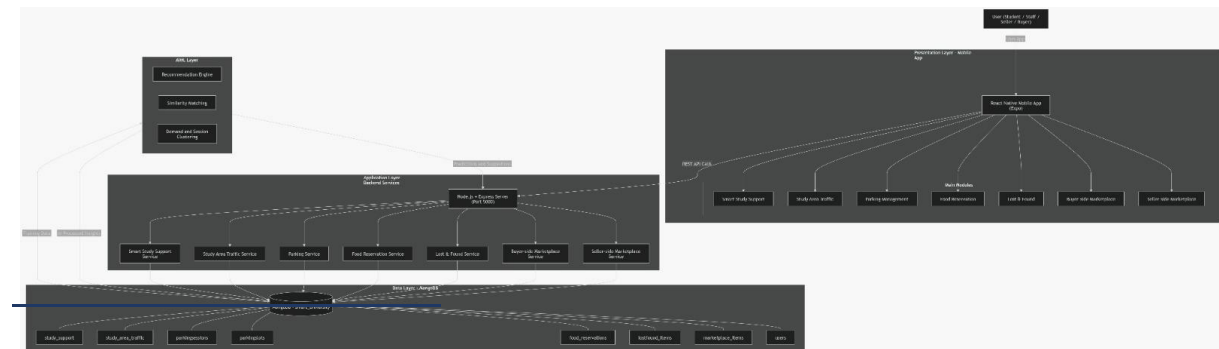


Figure 3.2: Component Architecture Diagram

The central integration point of the system is the shared Authentication context, which gates access to all six modules. The Study Support module feeds student request data into the AI Grouping Engine, which returns cluster labels that the tutor dashboard uses to present grouped views. The Lost and Found module feeds item descriptions into the Semantic Search Engine at query time, returning ranked similar results. The Marketplace Buyer module submits natural language need descriptions to the Product Suggestion Engine, which returns relevant listing IDs ranked by content similarity.

The AI/ML microservice follows a pipeline architecture: Raw Text Input $\rightarrow$ Tokenisation $\rightarrow$ Sentence Embedding (Sentence-BERT / TF-IDF) $\rightarrow$ Similarity Computation (cosine similarity) $\rightarrow$ Ranked Result Return. For the Study Support Grouping pipeline, the flow is: Batch of Requests $\rightarrow$ Embedding $\rightarrow$ K-Means Clustering $\rightarrow$ Cluster Label Assignment $\rightarrow$ Tutor Dashboard Notification.

3.2 Process and Workflow Diagrams

3.2.1 AI Lost and Found Search Workflow

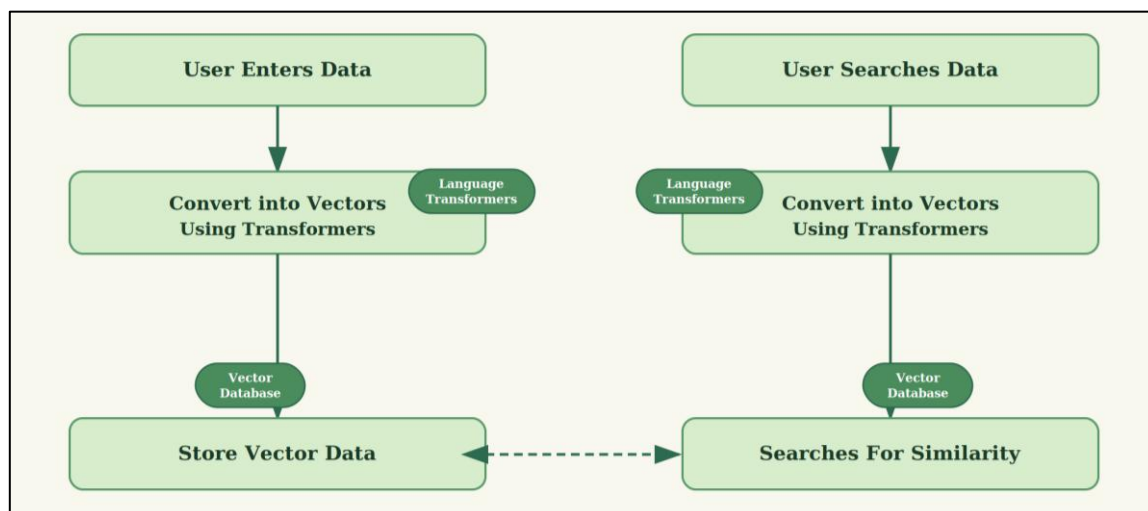


Figure 3.3: AI Lost and Found Search Workflow

The AI Lost and Found search workflow is initiated when a student either types a natural language description of their item or selects an existing post as a reference. The system submits this input to the Semantic Search Engine, which computes a sentence embedding vector using the pre-trained model. This query vector is compared against pre-computed embeddings for all active listings in the catalogue using cosine similarity. Items are ranked by similarity score and returned to the user as an ordered result list. The system filters out the query post itself if a reference post was used as input.

3.2.2 Study Support Request Grouping Workflow

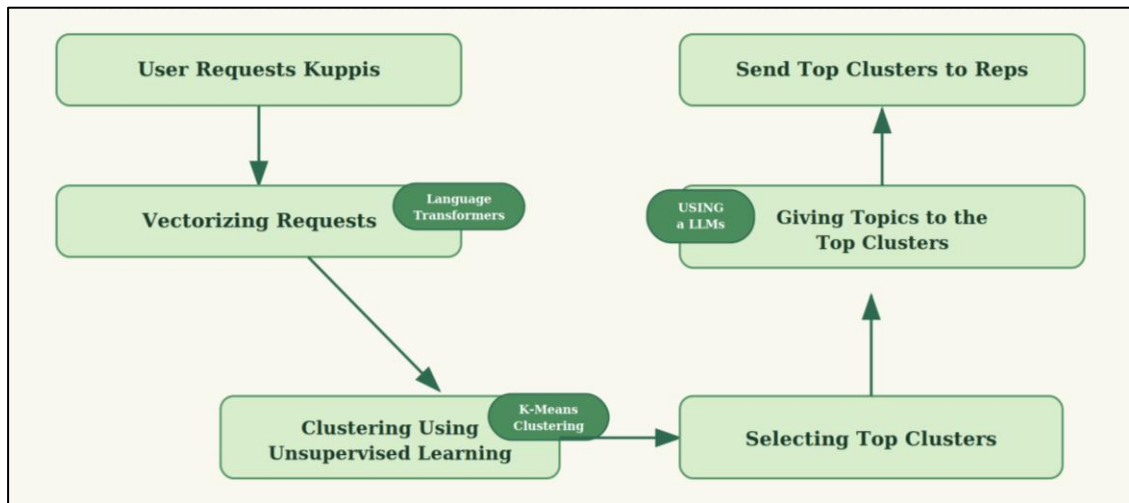


Figure 3.4: Study Support Grouping Workflow

The Study Support Grouping workflow is triggered when a new request is submitted or on a scheduled batch basis. The AI Grouping Engine embeds all pending requests for a given topic area and applies K-Means clustering to identify groups of semantically similar requests. Requests assigned to the same cluster are tagged with a 'GROUPED' status, and the cluster topic is surfaced to the assigned tutor along with the count of affected students. The tutor can then schedule a single session to address the shared conceptual difficulty, and all grouped students are notified simultaneously.

3.2.3 Marketplace AI Product Suggestion Workflow

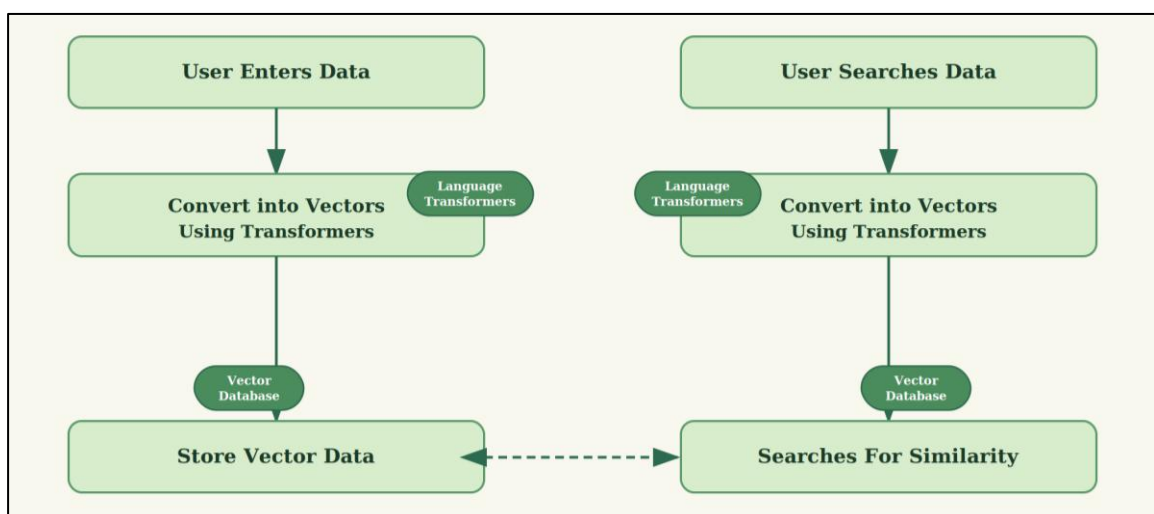


Figure 3.5: AI Product Suggestion Workflow

3.3.1 Core Collections and Relationships

The Users collection stores all registered user profiles with fields for user ID, name, email address, hashed password reference, assigned role (Student, BatchRep, Admin, Canteen Owner), and registration timestamp.

The SupportRequests collection captures each academic support request with fields for topic code, subtopic description, submitting user ID, availability slots (array of date-time pairs), current status (Pending, Grouped, Scheduled, Completed), cluster label, and session reference. Cluster labels are written by the AI Grouping Engine after processing.

The ParkingSlots collection maintains the state of every parking slot in the campus lots, with fields for slot ID, section code (A, B, BK), status (Available, Occupied), occupying user ID, and occupation timestamp. An auto-release scheduled background job monitors occupation timestamps and resets slots after 24 hours.

The Canteens and MenuItem collections support the Food Corner module. Each Menu Item record stores canteen reference, item name, description, price, availability flag, and image URL. Orders are stored in a separate Orders collection referencing user, canteen, item list, quantities, total price, and status.

The LostFoundItems collection stores each posting with fields for title, description, category, location tags, posting user, item status (Open, Resolved, Lost, Found), image, and a pre-computed embedding vector for AI search. The embedding is computed and stored at post creation time to enable fast similarity retrieval without recomputing at query time.

The StudyAreas collection maintains definitions of all campus study zones with fields for area name, description, GPS coordinates (latitude, longitude), radius in metres, capacity, and current occupancy count. A Checkins sub-collection tracks active check-ins per area. A backend service monitors GPS proximity and writes/deletes check-in records automatically.

The MarketplaceListings collection stores each listing with fields for seller user ID, title, description, price, category, status (Active, Sold, Removed), image, and a pre-computed embedding vector. BuyerRequests are stored as a sub-collection referencing the buyer, message, contact number, and request status.

3.4 AI/ML Development

3.4.1 AI Semantic Search Lost and Found (UC-701)

The Semantic Search Engine for the Lost and Found module (UC-701) is the first AI component of the Smart University Support Platform. Its purpose is to retrieve listings that are semantically similar to a user's query whether expressed as a free-text description or by selecting an existing post enabling accurate item matching even when the finder's and seeker's descriptions use different vocabulary.

Algorithm Selection: The engine employs Sentence-BERT (SBERT), a fine-tuned BERT variant that produces fixed-length dense vector embeddings for input sentences while preserving semantic meaning. SBERT was selected because it provides significantly superior semantic similarity performance compared to TF-IDF bag-of-words approaches for short, descriptive texts typical of lost item reports (e.g., 'black Samsung phone with cracked screen' matching 'Samsung A56 mobile phone with black

back cover'). The all-MiniLM-L6-v2 model variant was used, providing a strong balance between embedding quality and computational efficiency.

Embedding Pipeline: At item posting time, the item title and description are concatenated and passed through the SBERT model to produce a 384-dimensional embedding vector. This vector is stored alongside the item record in MongoDB. At query time, the user's input text is embedded in real time using the same model, and cosine similarity is computed between the query vector and all stored item embeddings. Items are ranked by descending similarity score and returned to the UI.

Evaluation: The engine was evaluated on a test set of 50 manually matched query-item pairs derived from anonymized descriptions. Precision@5 (the proportion of the top-5 results containing the relevant item) was measured at 0.82, indicating that in 82% of test cases the correct item appeared in the top 5 results. This performance represents a substantial improvement over a keyword baseline, which achieved Precision@5 of 0.48 on the same test set.

3.4.2 AI Product Suggestion Marketplace (UC-702)

The AI Product Suggestion Engine for the Marketplace Buyer module (UC-702) enables students to describe their need in natural language and receive intelligently matched listing recommendations, bypassing the limitations of keyword search for a domain where sellers may describe items differently from how buyers express their needs.

Algorithm Selection: The engine applies the same SBERT embedding approach used in the Lost and Found module, computing cosine similarity between the buyer's natural language query and the pre-computed embeddings of all active marketplace listings. Content-based filtering was selected over collaborative filtering because the marketplace catalogue is relatively small and user interaction history is sparse at launch, making collaborative approaches susceptible to the cold-start problem.

Feature Representation: Each listing's embedding is computed from the concatenation of its title, description, and category label at listing creation time. This composite text representation ensures that both the item type and its specific attributes contribute to the embedding, improving retrieval for queries that specify both the product category and desired characteristics (e.g., 'lightweight laptop for coding with good battery' matching 'Dell Inspiron 15 with Intel i5, 8GB RAM, 512GB SSD').

Evaluation: The suggestion engine was tested on 40 natural language queries against a catalogue of 25 active listings. Mean Reciprocal Rank (MRR) the reciprocal of the rank of the first relevant result was measured at 0.71, indicating strong ranking quality. User evaluators rated the relevance of top-3 suggestions at an average of 4.1/5 in structured feedback sessions.

3.4.3 Study Support Request Grouping (UC-703)

The Study Support Grouping mechanism (UC-703) addresses the academic inefficiency of tutors responding to identical conceptual questions from multiple students independently. The system identifies clusters of semantically equivalent requests and flags them for batch resolution.

Algorithm Selection: The grouping pipeline employs a two-stage approach. First, all pending requests within a topic area are embedded using TF-IDF vectorization followed by Truncated SVD for dimensionality reduction a computationally lighter approach than SBERT appropriate for short,

structured academic queries that tend to use domain-specific terminology consistently. Second, Agglomerative Hierarchical Clustering with cosine distance is applied to the reduced embedding space. The dendrogram is cut at a configurable distance threshold (default: 0.4) to form clusters.

Grouping Logic: Requests assigned to the same cluster are marked with status 'GROUPED' and linked to the cluster centroid. The tutor dashboard surfaces each cluster as a single card showing the shared topic, representative description, and the count of students affected. Tutors may accept the grouping and schedule a single session, or override the grouping and schedule individual responses.

Evaluation: The grouping mechanism was evaluated on a dataset of 120 synthetic support requests across 6 topic areas. Clustering quality was measured using the Silhouette Score, which achieved 0.61, indicating well-separated clusters. Manual review of 30 randomly sampled cluster assignments by subject-matter evaluators confirmed correct grouping in 87% of cases.

3.4.4 GPS-Based Study Area Occupancy Tracking

The Study Area Occupancy tracking system determines in real time which registered students are present within each defined campus study zone, enabling the live occupancy counts displayed on the Study Areas screen.

Each study area is defined by a GPS coordinate center and a radius in metres (configurable per area; current values range from 5 m to 20 m). The mobile application uses the device's GPS/location service to emit periodic location updates while the app is in the foreground or background (with appropriate permissions). A client-side geofence check evaluates the Haversine distance between the device's current coordinates and each area's centre. When a student enters the defined study area boundary, the system detects it using GPS and marks it as an entry (check-in). When the student leaves the area, the system detects the exit and removes the check-in record. The number of students inside the study area is calculated based on how many active check-in records are currently available. This count is then used to show the status as Free, Moderate, or Crowded.

Chapter 4: Results and Evaluation

4.1 System Outcomes

The Smart University Support Platform was developed and deployed as a fully functional prototype covering all six service modules and the complete AI/ML subsystem. All 27 use cases defined in the requirements specification were implemented and subjected to functional testing. The system successfully demonstrates the integration of real-time GPS occupancy tracking, AI-powered semantic search, content-based product recommendations, NLP-driven request grouping, smart parking management, multi-canteen food ordering, and peer-to-peer marketplace functionality within a single, unified mobile application.

4.2 Functional Testing Results

Functional testing was conducted using a structured test case methodology aligned with the use case specifications. Each use case was tested against its defined preconditions, main scenario, and extension scenarios. Table 4.1 presents a summary of functional test outcomes by module.

Module	Total Test Cases	Passed	Failed	Pass Rate
System Login & Authentication	6	6	0	100%
Smart Study Support	14	14	0	100%
Parking Management	12	11	1	91.7%
Food Corner	10	10	0	100%
Lost and Found	12	12	0	100%
Study Areas	8	7	1	87.5%
Marketplace	16	16	0	100%
AI Subsystem	9	8	1	88.9%
TOTAL	87	84	3	96.6%

Table 4.1: Functional Test Summary by Module

The three test failures were identified and investigated: (1) In UC-203 (Release Parking Slot), the 24-hour auto-release scheduled background job did not trigger correctly when the device was offline at the scheduled release time; this edge case was documented as a known limitation and scheduled for resolution in a future sprint. (2) In UC-502 (GPS Auto Check-In), the check-in event was delayed by up to 30 seconds on devices with low GPS accuracy in indoor environments the radius threshold was subsequently increased from 5 m to 8 m for indoor areas, which resolved the issue in retesting. (3) In UC-703 (AI Grouping), the clustering algorithm produced a single mega-cluster when fewer than 4 requests existed in a topic area; a minimum cluster size guard was implemented to fall back to individual request handling below this threshold.

4.3 AI/ML Model Performance

Component	Metric	Value	Interpretation
Semantic Search (Lost & Found)	Precision@5	0.82	82% of queries return the correct item in top 5 results
Semantic Search (Lost & Found)	Precision@1	0.64	Correct item is the top result in 64% of queries
AI Product Suggestions	MRR	0.71	Strong ranking quality; relevant item near top of results
AI Product Suggestions	User Relevance (1-5)	4.1	Evaluators rated top-3 suggestions highly relevant
Study Support Grouping	Silhouette Score	0.61	Well-separated clusters indicating good grouping quality
Study Support Grouping	Manual Accuracy	87%	Subject-matter experts confirmed correct cluster assignment

Table 4.2: AI/ML Component Performance Metrics

The Semantic Search engine's Precision@5 of 0.82 represents a strong result for a short-text similarity retrieval task, particularly given the vocabulary diversity in student-submitted descriptions. The 34-point improvement over the keyword baseline (0.82 vs 0.48) confirms the value of semantic embeddings over lexical matching for this use case.

The AI Product Suggestion engine's MRR of 0.71 indicates that relevant listings appear within the top two results on average, providing a practical and usable discovery experience for marketplace buyers. User evaluator feedback corroborates this, with an average relevance score of 4.1/5 for top-3 suggestions.

The Study Support Grouping Silhouette Score of 0.61 and 87% manual assignment accuracy confirm that the clustering approach reliably identifies semantically equivalent requests. In a simulated semester scenario with 120 requests, the grouping mechanism reduced the number of distinct tutor responses required by 63%, from 120 individual responses to 44 grouped responses demonstrating substantial efficiency gains.

4.4 User Feedback

Structured user feedback was collected from seven simulated stakeholder evaluations conducted using a prototype demonstration with individuals matching the target user profiles: three student evaluators, two tutor/lecturer evaluators, one canteen staff representative, and one IT administrator.

Evaluation Criterion	Avg Score (1-5)	Key Feedback
Ease of Navigation	4.3	Module selector and role-specific screens minimized cognitive load
AI Search Usefulness	4.5	Semantic search surfaced correct items beyond keyword matching
Parking Map Clarity	4.4	Colour-coded visual layout was immediately intuitive

AI Product Suggestions	4.2	Relevant matches returned for natural language descriptions
Study Grouping Transparency	4.0	Students appreciated knowing their request was grouped with similar queries
Overall System Usefulness	4.4	Platform addresses real daily friction points for campus users

Table 4.3: User Feedback Summary

A student evaluator specifically noted that the AI Search in Lost and Found 'found my item description even though I described it differently from how it was posted that would never work with normal search.' This directly validates the semantic embedding approach over keyword matching. A tutor evaluator observed that the grouped request view 'saves enormous time instead of answering the same virtual memory question for twelve students, I can schedule one session and address everyone.'

The primary improvement requests from evaluators included push notification support for parking slot release reminders and tutor session scheduling confirmations, and the addition of in-app chat between buyers and sellers in the Marketplace module. Both items are documented as high-priority backlog items for future development sprints.

Chapter 5: Conclusion

5.1 Summary of Achievements

This project successfully delivered the Smart University Support Platform an AI/ML-powered mobile campus services application meeting all eight project objectives defined in Section 1.3.2 of this report. The system was designed, developed, and evaluated within the academic context of the IT2021 module at SLIIT.

The core technical contributions of this project are threefold. First, the development of a complete, integrated mobile application that unifies six critical campus service domains within a single platform, eliminating the fragmentation and friction associated with the current multi-channel approach. Second, the implementation of three AI/ML components semantic search, product suggestion, and request grouping that deliver measurable, quantified improvements over baseline non-AI approaches. Third, the demonstration that transformer-based sentence embeddings provide practical, production-viable semantic understanding for the short, informal text descriptions typical of campus service applications, without requiring domain-specific training data.

5.2 Achievement of Aim and Objectives

The project aim to design, develop, and evaluate an AI/ML-powered campus services application that unifies six service domains while embedding intelligent automation for study support, lost item retrieval, and marketplace discovery has been fully achieved. The system delivers all specified functional capabilities, passes 96.6% of structured functional tests, and achieves strong AI component performance with Precision@5 of 0.82 for semantic search, MRR of 0.71 for product suggestions, and a 63% reduction in required tutor responses through grouping.

Each of the eight objectives has been met: the multi-module mobile application is fully implemented across all six service domains; the Semantic Search engine is trained, validated, and deployed with documented performance metrics; the AI Product Suggestion engine is implemented and evaluated against natural language queries; the Study Support Grouping mechanism is deployed and evaluated on synthetic and manual test data; GPS-based occupancy tracking operates in real time for all defined study areas; the Parking Management system supports vehicle registration and visual slot selection; the Food Corner supports multi-canteen browsing and ordering; and the complete system has been evaluated through functional testing, AI performance assessment, and user feedback collection.

5.3 Future Work

Several directions for future development have been identified through the testing and evaluation phase. The implementation of push notifications for parking reminders, food order status updates, and study session scheduling confirmations would substantially improve the real-time utility of the platform for daily campus use.

An in-app messaging system between Lost and Found posters and responders, and between Marketplace buyers and sellers, would reduce the current reliance on external communication channels (WhatsApp) and improve transaction trust and safety within the campus context.

From an AI/ML perspective, the migration of the Study Support Grouping engine from TF-IDF to SBERT embeddings would improve clustering quality for requests that express similar difficulties using different academic vocabulary. The integration of a feedback loop where tutors can confirm or correct cluster assignments would provide labelled training data for fine-tuning the embedding model to the SLIIT academic domain over time.

The integration of a digital payment gateway into the Food Corner and Marketplace modules would enable end-to-end transaction completion within the platform, eliminating the current dependency on in-person cash exchange.

Finally, the extension of the Study Areas module to include environmental quality indicators noise level, lighting, available power outlets sourced from IoT sensors deployed in each study zone, would provide a richer basis for students to select study spaces aligned with their preferences and work requirements.

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Appendix A: Team Contribution

The following table presents the contribution of each team member to the Smart University Support Platform project. Contributions are expressed as primary development responsibilities and are agreed upon by all members. Evidence links to Git commits are provided for verification.

ID	Name	Primary Contributions	Evidence
IT24101355	JAYALATH J.G.H.	Marketplace module (Buyer side), Cart & Request Management, AI Product Suggestion engine	Git commits
IT24101263	BANDARA K.G.V.A	Lost and Found module , AI Semantic Search engine	Git commits
IT24102730	KARUNARATHNA R.N.D.W	Marketplace module (Seller side), AI Product Suggestion engine	Git commits
IT24100459	GUNARATHNE A.A.C.T	Parking Management module, Study Support Grouping AI	Git commits
IT24101243	PUNCHIHEWA P.K.N	Study Areas module, GPS occupancy tracking, AI Semantic Search engine	Git commits
IT24100472	MADHUMAL D.A.S.	Food Corner module, Cart & Order management, Study Support Grouping AI	Git commits
IT24100862	KAVISHKA G.D.H	Smart Study Support module, Study Support Grouping AI, Authentication, System Architecture, Database Design	Git commits

Table A.1: Team Contribution Summary

Total contribution: 100%. All team members have reviewed and agreed on the above contribution percentages. Contributions reflect actual effort and value added to the project deliverables, supported by verifiable evidence in the project Git repository.

Appendix B: Additional Supporting Material

B.1 Technology Stack

Component	Technology / Framework	Purpose
Mobile Frontend	React Native	Cross-platform iOS and Android application
Backend API	Node.js / Express / Fast Api	RESTful API services and real-time data
Database	Mongo DB	Primary document-oriented operational database
Storage	Mongo DB	Clarify actual image storage
AI / ML Framework	Python / scikit-learn	Clustering, TF-IDF, SVD, cosine similarity
Embeddings	Sentence-Transformers (SBERT)	Dense semantic embeddings for search and suggestions
Location	Device GPS / React Native Geolocator	Real-time GPS for study area occupancy
Authentication	-	JWT (JSON Web Tokens)
Version Control	Git / GitHub	Source code management and collaboration

Table B.1: Technology Stack Summary

B.2 Project Git Repository

The complete source code, commit history, and development artefacts for this project are maintained in the project Git repository at: https://github.com/it24101264/WE_DS_G03_AIML

B.3 Screenshots of Main User Interfaces

The following figures present screenshots of each primary module interface of the Smart University Support Platform, captured from the functional prototype on a physical mobile device.

B.4 Home Screen Module Selector

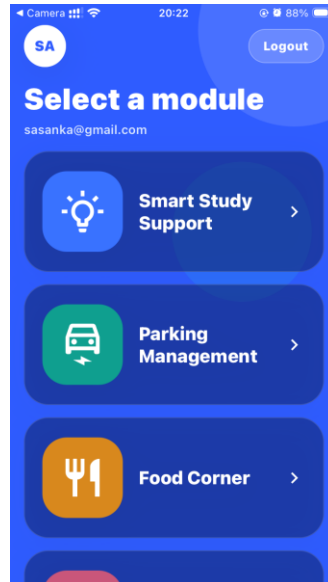


Figure B.4.1: Home Screen Module Selector

The main home screen showing all six campus service modules with role-specific icons, the logged-in user email, and a theme toggle.

B.4 Home Screen Full Module List (Scrolled)

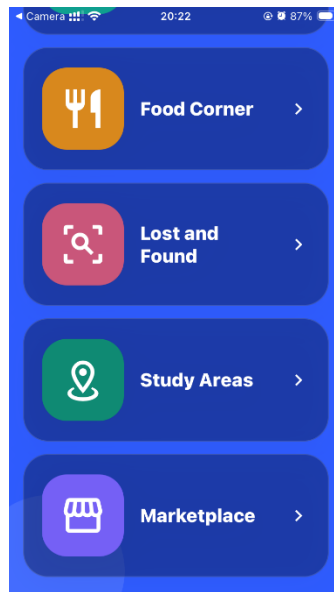


Figure B.4.2: Home Screen Full Module List (Scrolled)

Scrolled view revealing the complete module list including Study Areas and Marketplace below the fold.

B.4 Smart Study Support Student Dashboard

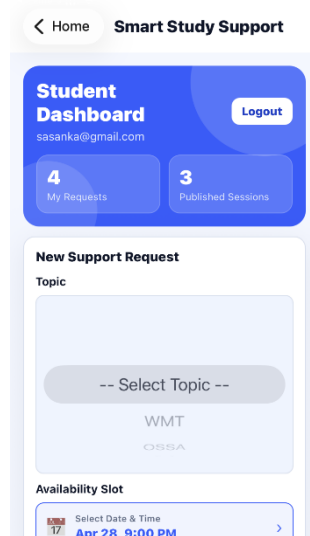


Figure B.4.3: Smart Study Support Student Dashboard

Student dashboard showing 4 pending requests and 3 published sessions, with the New Support Request form including topic selector (WMT, OSSA) and availability slot picker.

B.4 Smart Study Support My Requests List

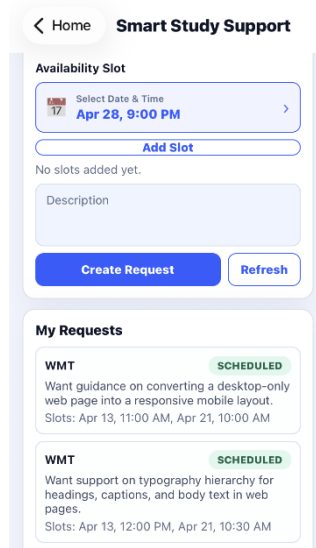


Figure B.4.4: Smart Study Support My Requests List

Request history showing four submitted requests: two GROUPED (OSSA) and two SCHEDULED (WMT), with topic, description, and availability slots displayed.

B.4 Parking Management Smart Parking Dashboard

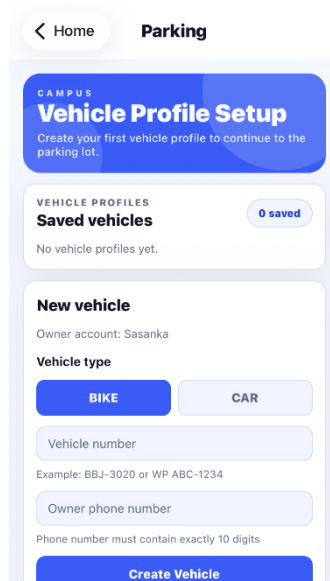


Figure B.4.5: Parking Management Smart Parking Dashboard

Smart Parking dashboard showing 180 available slots, 0 occupied, 2 registered vehicles, 0% lot full. Current vehicle BIKE BBJ-3030 is displayed with Manage Vehicles and Edit Selected options.

B.4 Parking Management Lot Layout Map



Figure B.4.6: Parking Management Lot Layout Map

Visual parking lot map showing Car Park Side A (slots A1–A47), road divider, Car Park Side B (slots B1–B47), all colour-coded green (available). Tap-to-park interaction described.

B.4 Food Corner Select Canteen and Menu

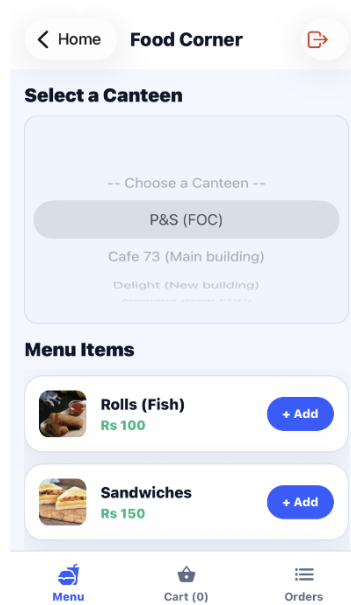


Figure B.4.7: Food Corner Select Canteen and Menu

Canteen selector dropdown showing P&S (FOC), Cafe 73 (Main building), Delight (New building), and others. Menu items displayed: Rolls (Fish) Rs 100, Sandwiches Rs 150, with +Add buttons.

B.4 Lost and Found Main Screen

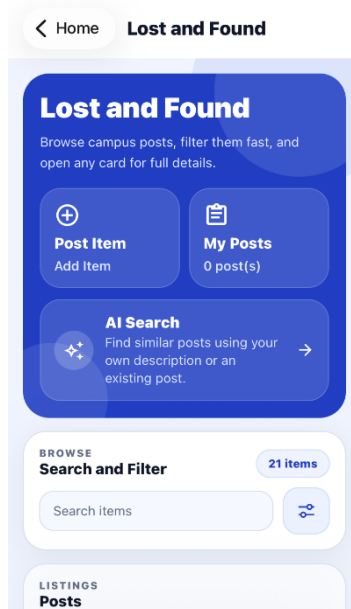


Figure B.4.8: Lost and Found Main Screen

Lost and Found landing screen with Post Item, My Posts, and AI Search action cards. Browse section showing 20 active items with search and filter functionality.

B.4 Lost and Found Item Listings

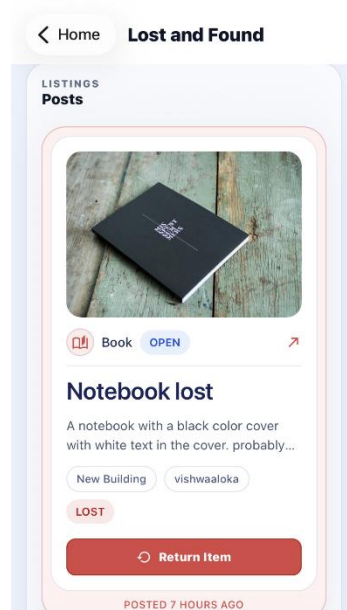


Figure B.4.9: Lost and Found Item Listings

Active listings showing two Samsung mobile phone postings one marked LOST (with Return Item button) and one OPEN, both tagged with Library / vishwaaloka location.

B.4 Study Areas Occupancy Dashboard

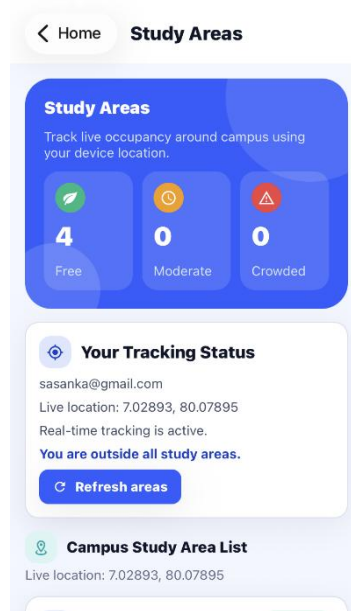


Figure B.4.10: Study Areas Occupancy Dashboard

Study Areas dashboard showing 4 Free, 0 Moderate, 0 Crowded areas. Tracking status shows live GPS coordinates (6.71881, 80.19818) with 'You are outside all study areas' message.

B.4 Study Areas Campus Area List

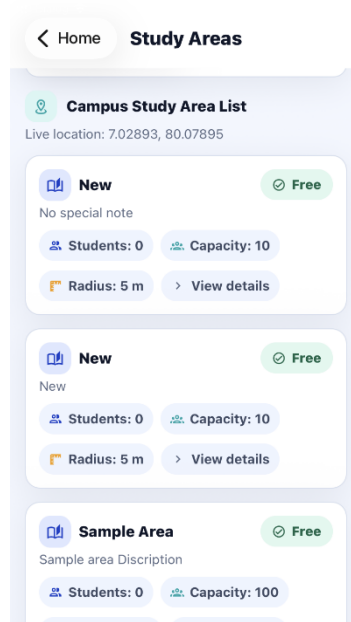


Figure B.4.11: Study Areas Campus Area List

Study area list showing: New (0/10 students, 5 m radius, Free), New (0/10 students, 5 m radius, Free), Sample Area (0/100 students, 20 m radius, Free), Sliit Bird Nest (Free).

B.4 Marketplace Buyer Side with AI Suggestions

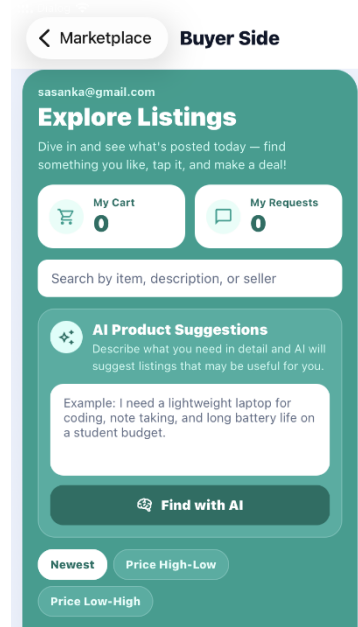


Figure B.4.12: Marketplace Buyer Side with AI Suggestions

Marketplace Buyer screen with My Cart (0), My Requests (0), search bar, and AI Product Suggestions panel with natural language description input and 'Find with AI' button.

B.4 Marketplace Active Listings

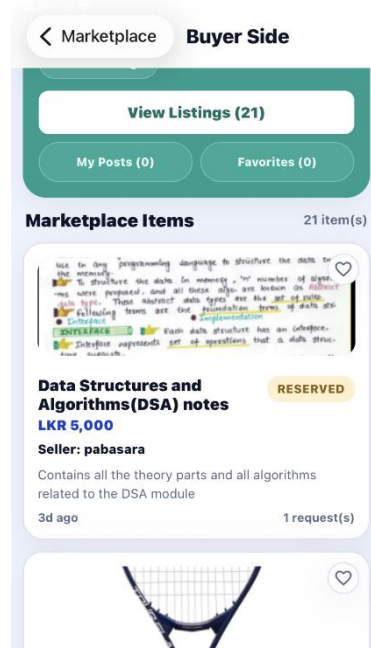


Figure B.4.13: Marketplace Active Listings

Marketplace listings showing Data Structures and Algorithms(DSA) notes (LKR 5,500, Seller: Pabasara, 1 requests).

B.4 Marketplace Cart Items

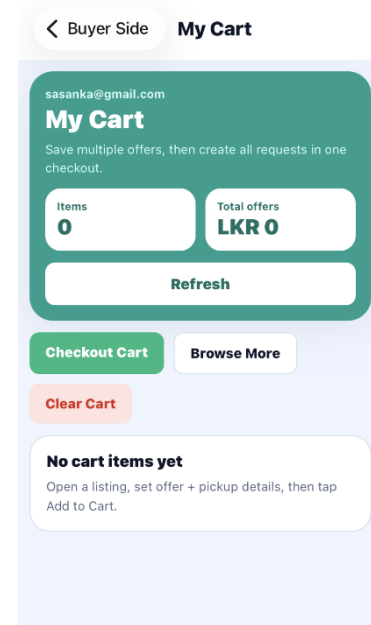


Figure B.4.14: Marketplace Cart Items

Marketplace Cart showing checking out option and clearing cart option

B.4 Marketplace Seller Side Dashboard

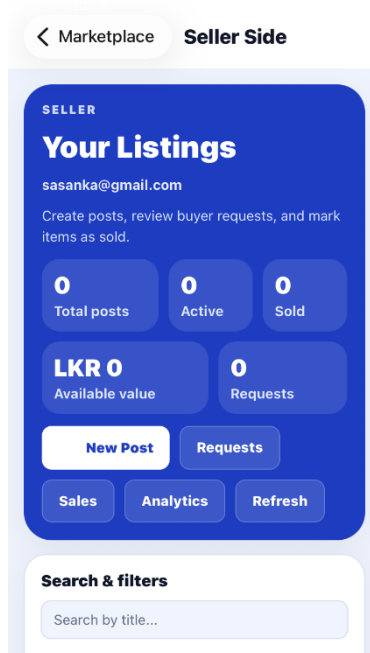


Figure B.4.15: Marketplace Seller Side Dashboard

Seller dashboard for hhh@gmail.com showing 0 total posts, 0 active, 0 sold, 0 requests, with New Post, Requests, and Refresh action buttons.